CCE-

UNIT-4

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* AVL Tree search path

* problem statement.
Write a c++ program to construct an AVL tree, searching for a specific key, and display the path followed during the search.

* Solution:
An AVL tree is a self-balancing Binary search tree. (BST). It maintains balance by performing rotations whenever necessary during insertion.

In this program:

* objectives:
- Insert elements into an AVL Tree.
- Maintain the balance of the tree using rotations.
- Accept a key from user to search.
- Display the path followed from the root to the key.
- Display the message if the is found or not found.
- To determine wheter the specified key is found or not found, in the tree.

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* Algorithm:

1) Start.
2) Create an empty AVL tree.
3) Read the no of elements.
4) Insert each element into the AVL Tree.
5) calculate the height & balance factor of each node.
6) Perform the required rotations to maintain AVL balance.
7) Read the key to be searched.
8) Start searching from the root node.
9) If the key matches the current node, display the search path and print "key found".
10) If the key is smaller, move to the left child.
11) If the key is greater, move to the right child.
12) Repeat the search until the key is found or a NULL node is reached.
13) If the key is not found, display "key not found".
14) Stop.

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* Flowchart:

